

BUBBLE FORMATION FROM A SUBMERGED SINGLE ORIFICE ACCOMPANIED BY PRESSURE FLUCTUATIONS IN GAS CHAMBER

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An improved two-stage model is proposed to predict bubble volume and to explain the phenomena of bubble formation when bubbles are formed from a submerged single orifice accompanied by pressure fluctuations in a gas chamber.

The following conclusions are reached by comparing the calculated results based on the proposed model with the experimental results:

- (1) The calculated bubble volumes agree well with the experimental results over a wide range of experimental conditions.
- (2) The change of bubble volume with bubbling time and the phenomena of pressure fluctuations in the gas chamber are well described by the model.

Introduction

Many studies have been made of the formation mechanism of bubbles generated from a single orifice submerged in a liquid, and it is known that the gas chamber volume beneath the orifice exerts considerable influence on the bubble volume¹⁰⁾.

For bubble formation accompanied by pressure fluctuation in the gas chamber, Khurana *et al.*³⁾, Kupferberg *et al.*⁵⁾ and MacCann *et al.*⁷⁾ proposed models consisting of two stages, that is, a first stage which corresponds to bubble expansion and a second for bubble expansion and/or uplift.

LaNauze and Harris⁶⁾ pointed out that the model of Khurana *et al.*³⁾ did not predict adequately the bubble volume change with bubbling time, because they did not treat bubble expansion dynamically. The equations derived by Kupferberg *et al.*⁵⁾ and MacCann *et al.*⁷⁾ are rather reasonable, but they did not consider the effect of viscosity. Hence, though the calculated results by their models predicted well the experimental results under a certain condition, agreement was still unsatisfactory for a wider range of variables, such as orifice diameter, gas chamber volume and gas flow rate.

The object of the present paper is to propose an improved bubble formation model which explains experimental observation better than do the previous models^{5,7)} when bubbles are formed at a submerged single orifice accompanied by pressure fluctuations in the gas chamber. The main point of the improve-

ment is the inclusion of the effect of viscosity in the equations which describe bubble formation. To examine the validity of the proposed model, the computed results of bubble size and of pressure fluctuations in the gas chamber based on the model are compared with the experimental results.

1. Model for Bubble Formation and Mathematical Formulation

Bubble formation consists of two periods, that is, the holding period and the formation period. When a bubble is detached from an orifice, the holding period starts for the next bubble. It ends when the pressure in the gas chamber reaches the sum of the average hydrostatic pressure and the surface tension pressure. During the holding period, there is only inflow of gas into the gas chamber.

Following the holding period is the bubble formation period, which consists of the following two stages:

(a) Expansion stage: The bubble grows while it remains at the orifice. This stage ends when the inertial force, surface tension and viscous drag force which act on the bubble are just balanced, so that the bubble begins to rise.

(b) Detachment stage: The bubble continues to grow while lifting up from the plate, but is still connected to the orifice by a small gas column, or neck. This stage terminates when the neck breaks off and the bubble separates. It is assumed that the neck breaks off when its length reaches a specified value.

The equations which describe the phenomena mentioned above are formulated in the following manner.

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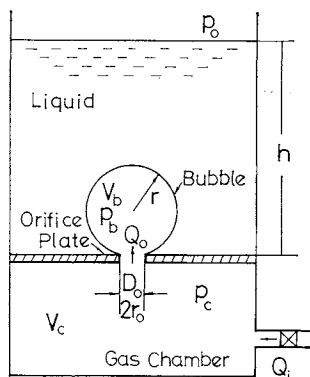


Fig. 1 Schematic diagram of bubble formation in expansion stage.

Throughout bubble formation it is assumed that the bubble grows spherically and that its motion is not affected by the presence of other bubbles.

For the relation between the difference of pressures in the gas chamber p_c and in the bubble p_b and the flow rate of gas Q_o , the following orifice equation is assumed to be applicable⁵⁾:

$$|p_c - p_b| = (Q_o/k_o)^2 \quad (1)$$

where k_o is an orifice constant. The situation under consideration is illustrated in Fig. 1.

Since Q_o is related to the rate of volume change of the bubble as

$$Q_o = \frac{dV_b}{dt} = 4\pi r^2 \frac{dr}{dt} \quad (2)$$

then Eq. (1) is rewritten as

$$|p_c - p_b| = \left(\frac{4\pi r^2}{k_o} \frac{dr}{dt} \right)^2 \quad (3)$$

Assuming adiabatic behavior in the chamber of volume V_c ⁵⁾, the change of pressure in the chamber during the expansion and detachment stages is expressed as

$$\frac{dp_c}{dt} = \frac{\gamma p_c}{V_c} (Q_g - Q_o) \quad (4)$$

where Q_g is the inflowing rate of gas into the chamber, and so is the average gas flow rate when the pressure of the upper stream of the gas chamber is large enough compared with the average pressure in the chamber. Substituting Eq. (2) into Eq. (4), we obtain

$$\frac{dp_c}{dt} = \frac{\gamma p_c}{V_c} \left(Q_g - 4\pi r^2 \frac{dr}{dt} \right) \quad (5)$$

When the pressure inside the bubble becomes greater than the sum of the average hydrostatic pressure and the surface tension pressure, the bubble expands by this pressure difference. The equation of the radial bubble expansion is written as

$$p_b - p_s = \phi \left(\frac{d^2 r}{dt^2}, \frac{dr}{dt}, \rho_l, \mu \right) + \frac{2\sigma}{r} \quad (6)$$

The average hydrostatic pressure p_s at the bubble

surface is approximately expressed as

$$p_s = p_o + \rho_l gh - \rho_l g(r - r_o) \quad (7)$$

where r_o is the radius of the orifice. In the right-hand side of the above equation, r_o is included to meet the initial condition that when $t=0$, $r=r_o$ and $p_s=p_o + \rho_l gh$.

For the ideal liquid, Kupferberg *et al.*⁵⁾ have derived the functional form of ϕ from the potential flow theory, but for a real liquid it is difficult to derive theoretically the concrete form of ϕ in Eq. (6). Hence, it is assumed here that ϕ is expressed as the product of the inertial term for the ideal liquid and a factor α which is a function of viscosity. Then ϕ becomes

$$\phi = \alpha \rho_l \left(r \frac{d^2 r}{dt^2} + \frac{3}{2} \left(\frac{dr}{dt} \right)^2 \right) \quad (8)$$

Other functional forms for ϕ may be adopted, and these are discussed briefly in Appendix.

Inserting Eq. (8) into Eq. (6), one obtains

$$p_b - p_s = \alpha \rho_l \left(r \frac{d^2 r}{dt^2} + \frac{3}{2} \left(\frac{dr}{dt} \right)^2 \right) + \frac{2\sigma}{r} \quad (9)$$

The equations for the expansion stage, Eqs. (3), (5) and (9), can be solved simultaneously under the initial conditions,

$$r = r_o, \quad dr/dt = 0 \quad \text{and} \quad p_c = p_o + \rho_l gh + 2\sigma/r_o$$

The termination of the expansion stage is the time when the following force balance equation holds:

$$\rho_l V_b g = \frac{d}{dt} \left(M' \frac{dr}{dt} \right) + 2\pi r_o \sigma + \frac{1}{2} C_D \rho_l \left(\frac{dr}{dt} \right)^2 \pi r^2 \quad (10)$$

where M' is the virtual mass for ascending motion of bubble and is assumed^{1,3)} to be $11\rho_l V_b/16$ according to Milne-Thomson⁸⁾.

The equation of motion for the bubble in the detachment stage is written as follows by considering the inertial force, the net bouyancy force and the viscous drag force on the bubble,

$$\frac{d}{dt} \left(M' \frac{dx}{dt} \right) = \rho_l V_b g - \frac{1}{2} C_D \rho_l \left(\frac{dx}{dt} \right)^2 \pi r^2 \quad (11)$$

where x is the vertical distance of center of bubble from the orifice plate.

The equations for the detachment stage, Eqs. (3), (5), (9) and (11), can be solved simultaneously. Initial conditions for the detachment stage are $r=r_{ef}$, $dr/dt=dr/dt|_{ef}$, $x=r_{ef}$, $dx/dt=dr/dt|_{ef}$ and $p_c=p_{cef}$, where subscript *ef* indicates the final value of the expansion stage.

As described later, the detachment condition is found experimentally to be approximately when the length of bubble neck becomes equal to the diameter of the orifice D_o :

$$x - r = D_o \quad (12)$$

As the drag coefficient C_D in Eqs. (10) and (11), it is appropriate to use the drag coefficient of bubble

sphere rather than that of solid sphere, so that C_D - Re relations of Tadaki *et al.*⁹⁾ and Kubota *et al.*⁴⁾ for a single bubble rising in purified liquids are used to build up the approximate equation of C_D for bubble sphere.

The simultaneous differential equations described above are solved numerically by Runge-Kutta-Gill method and the values of r , dr/dt , d^2r/dt^2 , x , V_b , p_e and p_b are obtained as a function of bubbling time t .

The value of parameter α appearing in Eq. (9) is determined by the following procedure:

Among the experimental results of the volume of bubbles successively generated from a single orifice submerged in a liquid, 12 data points are selected arbitrarily, whose experimental conditions, that is, orifice diameter, gas chamber volume and gas flow rate are different, and the calculated results of using Eq. (9) with trial values of parameter are compared with these 12 data. The appropriate value of parameter α is decided so as to minimize the average error between the calculated and experimental results.

The values of α thus obtained for distilled water, 20% methanol-water and 50% glycerine-water are expressed as the function of viscosity μ [poise] in this experimental range as

$$\alpha = 7.42 + 1.66 \log \mu \quad (13)$$

2. Experimental Apparatus and Procedure

A schematic diagram of the experimental apparatus is shown in Fig. 2. The flow rate of N_2 or air used was regulated by the manostat ① and measured by the capillary flow meter ②. Gas was fed into the gas chamber ⑤ through the orifice ④, and bubbles were formed successively by sending gas into bubble column ⑦ through an orifice ⑥. A bubble column of dimensions $120 \times 120 \times 1000$ (height) mm was constructed of brass with glass windows in front and back sides of the column. The orifices are brass plates 0.50 mm thick, each of which has a single hole of 0.199, 0.160, 0.111 or 0.071 cm in diameter at the center, and their orifice constants were determined experimentally. The liquid height was 30 cm. The gas chamber volume was varied from 20 to 205 cm^3 by changing the liquid level in the chamber.

Pressure fluctuation in the gas chamber during bubble formation was detected by the pre-calibrated micromanometer (bellows-type) ⑧ and recorded by an electromagnetic oscilloscope or by a cathode ray oscilloscope.

By detecting the change of intensity of light due to bubble formation at the orifice with a phototransistor, the time required to form ten bubbles was determined ten times for each run and an average was obtained, from which the frequency of bubble formation per second was calculated. The volume of a single bub-

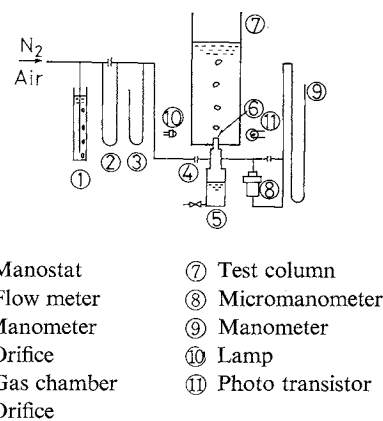


Fig. 2 Experimental apparatus

Table 1 Physical properties of liquids

Liquid	Temp. [°C]	ρ_l [g/cm ³]	μ [c.p.]	σ [dyne/cm]
Distilled water	26.0	0.997	0.874	71.8
50wt.% Glycerine in water	25.0	1.121	3.89	70.3
20wt.% Metahnl in water	25.0	0.966	1.33	48.9

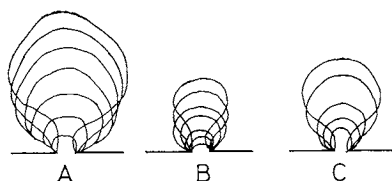
ble was calculated by dividing gas flow rate by the frequency of bubble formation based on the assumption that bubbles of the same size were formed.

To study the phenomena of bubble formation in detail, the bubbles forming at the orifice were photographed by a high-speed camera (Himac 16 M) for 8 conditions which are considered to be representative in the experimental range mentioned later. For the specified conditions, the oscilloscope record representing the pressure fluctuation in the gas chamber and the bubbles forming at the orifice were photographed simultaneously on high-speed film.

The films obtained were enlarged about 2.5 times the size of the actual bubble with a film analyzer. Bubble shapes during formation were traced and the relation between bubble volume and time was obtained. The bubble volume was calculated by Simpson's rule assuming that the bubble is a body of revolution. These analyses were conducted for five bubbles in each run.

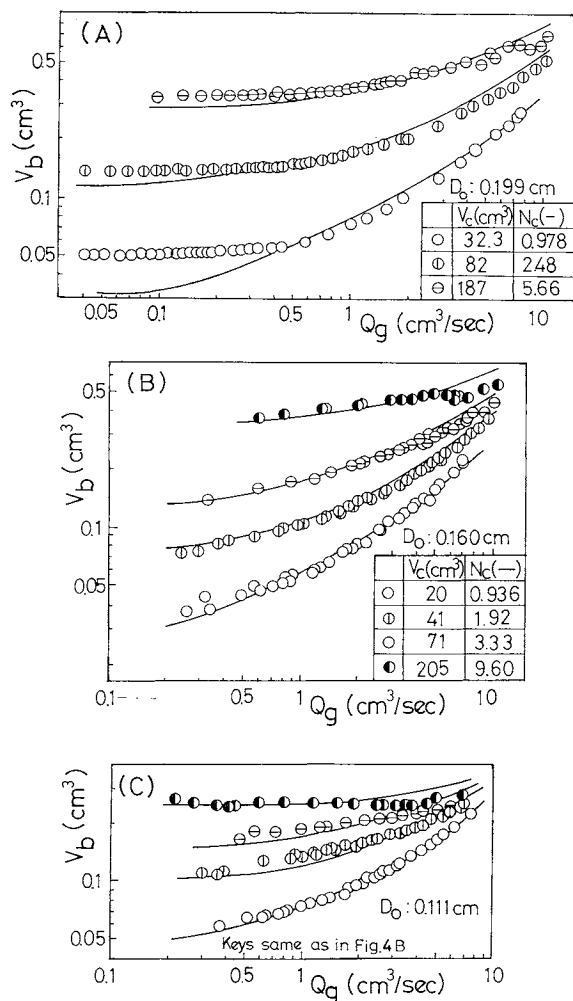
With reference to the experimental results reported by Tadaki *et al.*¹⁰⁾ that the bubble volume V_b depends on V_e in the range $1 < N_e < 9$, where N_e is the dimensionless parameter for gas chamber, experiments in this study were carried out mainly in this range. The experimental range for N_w , the dimensionless parameter for gas flow rate, was $0.3 < N_w < 14$, where bubbles are formed uniformly¹⁰⁾.

The liquids used were distilled water, 50 wt.% glycerine-water and 20 wt.% methanol-water. Their physical properties were measured and are shown in Table 1.



Experimental conditions are shown in Table 2.

Fig. 3 Changes of bubble shapes during bubble formation



Solid lines show the calculated results.

Fig. 4 Relations between V_b and Q_g with V_c as a parameter in distilled water

3. Results and Discussion

3.1 Bubble shape

Figure 3 shows traces of typical bubble shapes during bubble formation under the conditions shown in Table 2. As shown in the figure, the semispherical bubble grows to roughly spherical shape while remaining at the orifice. Then the bubble begins to lift up while changing its volume, and finally the neck breaks off when the bubble separates from the plate. These sequences of bubble formation show very well the two different stages before-mentioned, that is, the

Table 2 Conditions of bubble formation

No	D _o [cm]	V _c [cm³]	Q _g [cm³/sec]
A	0.199	187	5.52
B	0.199	32.3	0.692
C	0.111	32.3	1.30

expansion and detachment stages, and they are very similar to the observation of Kupferberg *et al.*⁵⁾

The lengths of the bubble neck just before bubble separation were measured and were nearly equal to the diameter of the orifices used. This result was also confirmed by Ishihara *et al.*²⁾ and gives the final condition of the detachment stage as written in Eq. (12).

3.2 Bubble volume

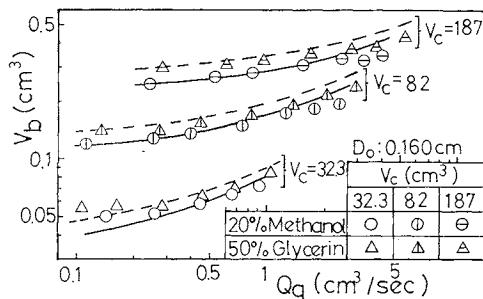
Figures 4 (A), (B), (C) show the relations between gas bubble volume V_b and gas flow rate Q_g in water with gas chamber volume V_c as a parameter, where orifice diameters are 0.199, 0.160 and 0.111 cm, respectively. It is seen that gas bubble volume increases with increase of chamber volume and gas flow rate. This tendency becomes remarkable as orifice diameter increases.

The results of calculation using the model described in section 2 are shown in the same figures. The calculated curves fit adequately over a wide range of experimental conditions.

But the calculated bubble volumes are smaller than the experimental ones at small gas flow rates as shown in Fig. 4 (A). It is found from the analyses of the results of calculation that the most likely reason for the discrepancy is that the detachment condition, Eq. (12), is not satisfied. The bubble leaves the orifice plate during the decrease of the bubble volume in this range, as will be mentioned in a later section (cf. condition B in Fig. 6). Hence, when the bubble separates before the length of bubble neck becomes equal to the diameter of the orifice, the experimental bubble volumes are larger than the calculated ones.

On the other hand, when the gas flow rate and the gas chamber volume become large, the calculated bubble volumes become larger than the experimental ones. This may be due to the wake of the previous bubble, which accelerates the upward motion of the bubble in the detachment stage. Thus, the bubble separates earlier and its volume become smaller than the calculation predicts.

To investigate the effect of the physical properties of liquids on bubble volume, experiments were conducted for 20 wt. % methanol-water and 50 wt. % glycerine-water solutions as shown in Table 1. Figure 5 shows the experimental results of the relation between bubble volume and gas flow rate in 20 wt. % methanol-water and 50 wt. % glycerine-water solutions together with the results of calculation. The calculated curves



Solid and broken lines show the calculated results for 20% methanol-water and for 50% glycerine-water solutions, respectively.

Fig. 5 Relations between V_b and Q_g for 20% methanol-water and for 50% glycerine-water solutions

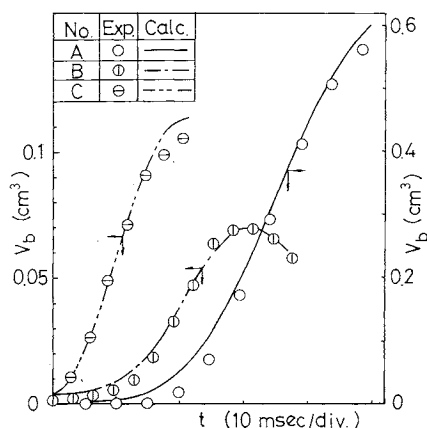


Fig. 6 Changes of bubble volume with time under the conditions shown in Table 2

fit generally well in this experimental range.

3.3 Bubble growth and pressure fluctuation in the gas chamber

To assess the adequacy of this model of bubble formation, it is important not only that prediction of final bubble volume is possible but also that the experimental results of the changes of bubble volume and pressure in the gas chamber with time are well described by the equations presented in section 2.

Figures 6 and 7 show examples of bubble volume and pressure in the gas chamber, respectively, as a function of growing time for the conditions shown in Table 2. In these figures, the experimental points are taken from the analyses of high-speed films and the curved lines show the calculated results from the model. Agreement between the experimental points and the calculated curves is fairly good.

The data points and curves presented in Figs. 6 and 7 show typical patterns of bubble formation. In conditions A and C, the bubble separates approximately when the bubble volume is maximum and the pressure in the gas chamber is minimum, and these phenomena were also observed by Kupferberg *et al.*⁵⁾. When the gas flow rate is small, as shown by condition

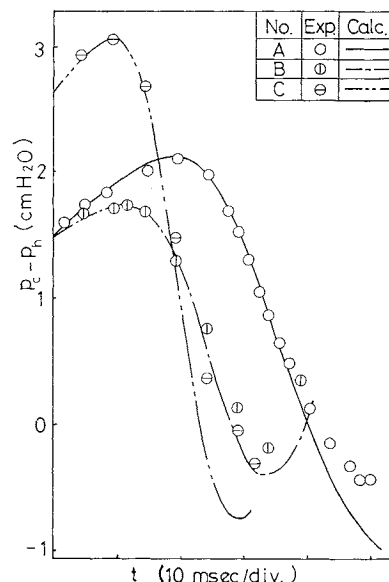


Fig. 7 Pressure fluctuations in the gas chamber during bubble formation under the conditions shown in Table 2

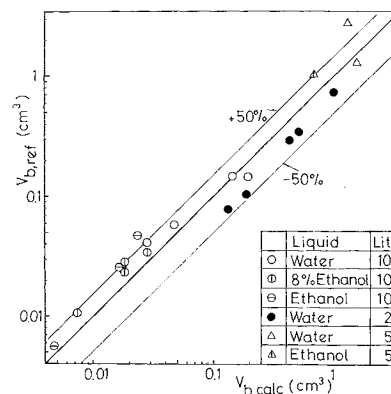


Fig. 8 Comparison of bubble volumes obtained from reference data and this calculation

B, the bubble separates after the maximum volume and the minimum pressure in the gas chamber are reached, as observed by Ishihara *et al.*²⁾.

3.4 Comparison with other researchers' results

The experimental results of bubble volume, $V_{b,ref}$ in water, 8% ethanol-water and ethanol by Tadaki *et al.*¹⁰⁾, those in water by Ishihara *et al.*²⁾ and those in water and ethanol by Kupferberg *et al.*⁵⁾, and the calculated results of bubble volume, $V_{b,calc}$, by this model are plotted in **Fig. 8**. These twenty data are selected over a wide range of experimental conditions, $1 < N_c < 9$ and $0.13 < N_w < 15$. The relation between $V_{b,ref}$ and $V_{b,calc}$ are correlated approximately within an accuracy of $\pm 50\%$.

Conclusion

To predict bubble volume and to describe the phenomena of bubble formation when bubbles form at a submerged single orifice in the presence of pressure

fluctuation in the gas chamber, a semitheoretical bubble formation model was proposed.

As for bubble volume, agreement between the results of calculation by this model and experimental values was fairly good over a wide range of experimental conditions. Also bubble growth and pressure fluctuation in the gas chamber were described rather well by the model.

The calculated results of bubble volume agreed generally well with the various existing data, so that it may be concluded that the presented model is practically useful in estimating the volume of bubbles formed at a submerged single orifice accompanied by pressure fluctuations in the gas chamber.

Appendix

In addition to Eq. (8), other functional forms for ϕ in the equation of bubble expansion, that is, Eq. (6), were examined.

The inertial term may be written generally in the form of $(1/A_b)(d/dt)(M_v(dr/dt))$, where A_b is the surface area of the bubble and M_v is the virtual mass of the bubble. Assuming that the virtual mass of the bubble, M_v , is proportional to the bubble volume:

$$\phi = \beta_1 \rho_l \left(\frac{r}{3} \frac{d^2 r}{dt^2} + \left(\frac{dr}{dt} \right)^2 \right) \quad (\text{A-1})$$

and if M_v is proportional to the bubble surface area A_b :

$$\phi = \beta_2 \rho_l \left(\frac{d^2 r}{dt^2} + \frac{2}{r} \left(\frac{dr}{dt} \right)^2 \right) \quad (\text{A-2})$$

Though various functional forms may be considered as the trial function for ϕ , here only the following simpler one was considered:

$$\phi = \beta_3 \rho_l \frac{d^2 r}{dt^2} \quad (\text{A-3})$$

The values of parameters β_1 , β_2 and β_3 appearing in Eqs. (A-1), (A-2) and (A-3) were determined in the same manner as in the case of α , appearing in Eq. (8). For distilled water, $\beta_1=12$, $\beta_2=0.8$ and $\beta_3=0.8$, respectively.

These four cases for ϕ were investigated and compared with the experimental results. As for bubble volume, agreement between the results of calculation by using Eq. (A-3) and the experimental results was best among the four cases. But agreement between calculated and experimental results on the bubble growth curve and the pressure change in the gas chamber was more excellent when Eq. (8) was used. The other two cases, that is, Eqs. (A-1) and (A-2), gave poorer results.

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Nomenclature

A_b	= surface area of bubble	[cm ²]
C_D	= drag coefficient of bubble	[—]
D_o	= orifice diameter	[cm]
Fr	= Froude number, U_o^2/gD_o	[—]
g	= gravitational acceleration	[cm/sec ²]
h	= liquid height	[cm]
k_o	= orifice constant	[(cm ³ /sec)/(dyne/cm ²) ^{0.5}]

M_v	= virtual mass of bubble	[g]
M'	= virtual mass of bubble in Eqs. (10) and (11)	[g]
N_c	= dimensionless parameter, $4V_c g \rho_l / \pi D_o^2 p_h$	[—]
N_w	= dimensionless parameter, $We/Fr^{0.5}$	[—]
p_b	= pressure in bubble	[dyne/cm ²]
p_c	= pressure in gas chamber	[dyne/cm ²]
p_h	= hydrostatic pressure at orifice plate, $p_o + \rho_l g h$	[dyne/cm ²]
p_s	= average hydrostatic pressure on the bubble surface in Eq. (7)	[dyne/cm ²]
p_o	= atmospheric pressure	[dyne/cm ²]
Q_g	= average gas flow rate	[cm ³ /sec]
Q_o	= outflowing rate of gas from gas chamber	[cm ³ /sec]
r	= equivalent spherical radius of bubble	[cm]
r_o	= orifice radius	[cm]
Re	= Reynolds number, $2r v \rho_l / \mu$	[—]
t	= bubbling time	[sec]
U_o	= gas velocity through orifice	[cm/sec]
V_b	= bubble volume	[cm ³]
V_c	= gas chamber volume	[cm ³]
v	= velocity of bubble center	[cm/sec]
We	= Weber number, $D_o U_o^2 \rho_l / \sigma$	[—]
x	= vertical distance of center of bubble from the orifice plate	[cm]
α	= parameter in Eq. (8)	
$\beta_1, \beta_2, \beta_3$	= parameters in Eqs. (A-1), (A-2) and (A-3)	[—]
γ	= specific heat ratio	[—]
μ	= liquid viscosity	[g/(cm·sec)]
ρ_l	= liquid density	[g/cm ³]
σ	= liquid surface tension	[dyne/cm]
ϕ	= function of inertial and viscous forces in Eq. (6)	

<Subscripts>

calc	= calculated value from the model
ef	= final value of the expansion stage
ref	= reference data

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